

# Digital Signal Analysis and Processing Numerical Companion

CT 704 (BEI/BCT, IV-I) / EX 753 (BEX, IV-II) · TU, IOE · tiers over 45 papers, 2066–2082 BS

## Ch 5 • FIR Filter Design

### Numerical Problems & Solutions

**47 of the 70 questions are one design calculation.** The other 23 are the Remez algorithm, which asks for a flowchart and never for a number: that is \$5.5 of the theory, not here. Every design paper is below.

#### The recipe, used in §1, §2 and §3

1. **Convert the spec** to  $\omega$  in radians per sample if it is given in Hz ( $\omega = 2\pi f/f_s$ ) or rad/s ( $\omega = \Omega/f_s$ ).
2. **Window** from the stopband attenuation  $A_s$ , taking the cheapest that clears it. If the paper gives  $\delta_s$ , first  $A_s = -20 \log_{10} \delta_s$ .
3. **Length**  $M = c\pi/\Delta\omega$  with  $c$  from the same table, then round up to the next **odd**  $N$ .
4. **Cut-off**  $\omega_c = \frac{\omega_p + \omega_s}{2}$ , the middle of the transition band.
5. **Coefficients**  $h[n] = h_d[n]w[n]$ , with  $h_d[n] = \frac{\sin(\omega_c(n-\alpha))}{\pi(n-\alpha)}$ ,  $\alpha = \frac{N-1}{2}$ , and the centre tap  $h_d[\alpha] = \omega_c/\pi$  done separately.

#### The table, again, because everything turns on it

| Window      | Stop band | Transition |
|-------------|-----------|------------|
| Rectangular | 21 dB     | $1.8\pi/M$ |
| Bartlett    | 25 dB     | $6.1\pi/M$ |
| Hann        | 44 dB     | $6.2\pi/M$ |
| Hamming     | 53 dB     | $6.6\pi/M$ |
| Blackman    | 74 dB     | $11\pi/M$  |

#### Checks that cost nothing

1. The centre tap is always  $\omega_c/\pi$  exactly. **If it is not,  $\omega_c$  is wrong.**
2.  $h[n]$  must be **symmetric** about  $\alpha$ : compute one half and mirror it.
3. A **Hann**-windowed filter has  $h[0] = h[N-1] = 0$  exactly, because the window is zero there. So "the first three coefficients" begins with a zero.
4.  $N$  odd, always, so  $\alpha$  is an integer.

### 1 Fixed-window low-pass designs

**TOP 13/45** 79 Ba **5+3**, 71 Ch, 69 Ch **8**, 74 Ash **2+6**, 79 Ch **8**, 75 Bh **6**, 82 Ba **2+6**, 67 Mng **8**, 70 Asa **9**, 69 Bh **8**, 68 Bh **9**, 76 Ch **6**, 81 Ch **2+6**, 73 Ma **4+5**, 82 Bh **10**

**QUESTION AS PRINTED** 2079 Chaitra · 2075 Bhadra · [8] [6]

Design a low pass FIR filter using a suitable window for  $\omega_p = 0.24\pi$ ,  $\omega_s = 0.54\pi$  and stopband attenuation  $\alpha_s = 42.2$  dB. Find the first three coefficients of the impulse response.

#### Step 1: the two lookups

- $A_s = 42.2$  dB. Hann gives **44 dB**, which clears it; Bartlett's 25 dB does not. **So Hann**, and Hamming would be over-design.
- $\Delta\omega = 0.54\pi - 0.24\pi = 0.30\pi$ .

$$M = \frac{6.2\pi}{\Delta\omega} = \frac{6.2\pi}{0.30\pi} = 20.67 \Rightarrow M = 21, \quad \boxed{N = 23}$$

rounding up, then up again to odd.

$$\alpha = \frac{N-1}{2} = \mathbf{11}.$$

#### Step 2: the cut-off, in the middle

$$\omega_c = \frac{0.24\pi + 0.54\pi}{2} = \mathbf{0.39\pi} = 1.2252 \text{ rad/sample}$$

$$h_d[n] = \frac{\sin(0.39\pi(n-11))}{\pi(n-11)}, \quad h_d[11] = \frac{\omega_c}{\pi} = \mathbf{0.39}$$

**Step 3: the first three coefficients**

- Hann window,  $w[n] = 0.5 - 0.5 \cos \frac{2\pi n}{22}$ :

$$w[0] = 0, \quad w[1] = 0.020254, \quad w[2] = 0.079373$$

$$h_d[0] = 0.022865, \quad h_d[1] = -0.009836, \quad h_d[2] = -0.035350$$

$$h[0] = 0, \quad h[1] = -0.000199, \quad h[2] = -0.002806$$

- $h[0] = 0$  is not a mistake: the Hann window is exactly zero at both ends, so the first and last taps of the filter vanish. Say so, because it looks like an error otherwise.
- The rest follows by symmetry,  $h[22 - n] = h[n]$ , with the centre tap  $h[11] = 0.39$ .

**Every other fixed-window design**

| Papers              | Spec                                                                                                                          | $A_s$   | Window           | $N$ | $\omega_c$ |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------|---------|------------------|-----|------------|
| 79 Ba               | $\omega_p = 0.2\pi, \omega_s = 0.45\pi$                                                                                       | 51 dB   | Hamming          | 29  | $0.325\pi$ |
| 71 Ch, 69 Ch        | $\omega_p = 0.3\pi, \omega_s = 0.5\pi$                                                                                        | 40 dB   | Hann             | 33  | $0.4\pi$   |
| 74 Ash              | $\omega_p = 0.2\pi, \omega_s = 0.5\pi$                                                                                        | 41 dB   | Hann             | 23  | $0.35\pi$  |
| 79 Ch, 75 Bh        | $\omega_p = 0.24\pi, \omega_s = 0.54\pi$                                                                                      | 42.2 dB | Hann             | 23  | $0.39\pi$  |
| 67 Mng              | $\omega_p = 0.35\pi, \omega_s = 0.45\pi$                                                                                      | 54 dB   | Blackman         | 111 | $0.4\pi$   |
| 76 Ch, 81 Ch, 73 Ma | $\delta_p = 0.101, \delta_s = 0.01$                                                                                           | 40 dB   | Hann             | 33  | $0.3\pi$   |
| 82 Bh               | $\omega_p = 0.2\pi, \omega_s = 0.4\pi$<br>2 kHz, 5 kHz at $f_s = 20$ kHz                                                      | 42 dB   | Hann             | 23  | $0.35\pi$  |
| 82 Ba               | $(\omega_p = 0.2\pi, \omega_s = 0.5\pi)$<br>−3 dB at $30\pi, 50$ dB at $45\pi$<br>rad/s, $f_s = 100$ Hz ( $0.3\pi, 0.45\pi$ ) | 50 dB   | Hamming          | 45  | $0.375\pi$ |
| 70 Asa              | Hann named, $0.25\pi, 0.35\pi$ ,<br>main lobe $8\pi/M$                                                                        | —       | Hann (given)     | 81  | $0.3\pi$   |
| 69 Bh               | Hann named, $0.25\pi, 0.3\pi$                                                                                                 | —       | Hann (given)     | 125 | $0.275\pi$ |
| 68 Bh               | Blackman named, $0.24\pi, 0.34\pi$ ,<br>main lobe $12\pi/M$                                                                   | —       | Blackman (given) | 121 | $0.29\pi$  |

**67 Mng is the marginal case, and that is the question** It asks for "at least 54 dB". Hamming gives 53, one decibel short, so it is not allowed and Blackman is forced. The cost is brutal: Hamming would have needed  $M = 6.6\pi/0.1\pi = 66$ , so  $N = 67$ , while Blackman needs  $M = 11\pi/0.1\pi = 110$  and  $N = 111$ . **One decibel of specification nearly doubles the filter.** That step, not a slope, is the whole point of the window table, and saying so is worth more than the arithmetic.

**70 Asa and 68 Bh hand you a main-lobe width instead of a transition figure** They say "main lobe width  $8\pi/M$ " (Hann) and " $12\pi/M$ " (Blackman) and expect you to use *that* for the length:  $M = 8\pi/\Delta\omega$  and  $12\pi/\Delta\omega$  rather than the exact 6.2 and 11. For 70 Asa,  $\Delta\omega = 0.1\pi$  gives  $M = 80$ , so  $N = 81$ ; for 68 Bh,  $\Delta\omega = 0.1\pi$  gives  $M = 120$ , so  $N = 121$ . **Use the constant the paper gives you**, and note in one line that the exact figures would give 63 and 111. Both conventions are in the textbooks and the marks follow the paper.

**2 Short designs with the window named**

PIN 3/45 80 Bh 10, 73 Bh 2+7, 76 Bh 1+2+7 · **no window choice and no length choice**: both are given

QUESTION AS PRINTED 2080 Bhadra · [10]

Design a symmetric linear-phase FIR low pass filter whose desired frequency response is  $H_d(\omega) = e^{-j\omega\tau}$  for  $|\omega| \leq \omega_c$  and 0 elsewhere. The length of the filter should be 7 and  $\omega_c = 1$  rad/sample. Use the Hanning window.

- $N = 7$ , so  $\alpha = \tau = \frac{N-1}{2} = 3$ , and the design is just "evaluate seven numbers twice and multiply".

$$h_d[n] = \frac{\sin(1 \cdot (n-3))}{\pi(n-3)}, \quad h_d[3] = \frac{1}{\pi} = 0.31831$$

$$w_{\text{Hann}}[n] = 0.5 - 0.5 \cos \frac{2\pi n}{6}$$

| $n$             | 0       | 1       | 2       | 3       | 4       | 5       | 6       |
|-----------------|---------|---------|---------|---------|---------|---------|---------|
| $h_d[n]$        | 0.01497 | 0.14472 | 0.26785 | 0.31831 | 0.26785 | 0.14472 | 0.01497 |
| Hann $w[n]$     | 0       | 0.25    | 0.75    | 1       | 0.75    | 0.25    | 0       |
| $h[n]$          | 0       | 0.03618 | 0.20089 | 0.31831 | 0.20089 | 0.03618 | 0       |
| Blackman $w[n]$ | 0       | 0.13    | 0.63    | 1       | 0.63    | 0.13    | 0       |
| $h[n]$          | 0       | 0.01881 | 0.16874 | 0.31831 | 0.16874 | 0.01881 | 0       |

- **73 Bh** is the same question with Hann, and **76 Bh** the same with Blackman, so the last two rows are that answer.
- Both windows kill the end taps, so a "7-tap" filter really has **5 non-zero coefficients**. **Worth one line:** it is why Hann and Blackman waste two taps on short filters, and why the rectangular window is not always wrong.
- Every row is symmetric about  $n = 3$ , confirming linear phase with a group delay of exactly 3 samples.

73 Bh and 76 Bh also ask "what is windowing, list the key points". Answer in four: **truncation is unavoidable** because  $h_d$  is infinite; truncating abruptly causes **Gibbs ripple**; a tapered window trades **ripple for transition width**; and the window is chosen by **attenuation**, the length by **transition width**.

### 3 Kaiser window designs

**TOP 13/45** 81 Ba **10**, 81 Bh **8**, 80 Bh **2+2+2**, 80 Ba **2+8**, 79 Bh **2+8**, 78 Bh **8**, 74 Ch **9+3**, 75 Ash **8+4**, 72 Ka **6+8**, 76 Ash **2+4+4**, 73 Ch **8**, 78 Ch **8**, 66 Ma

**QUESTION AS PRINTED** 2081 Baishakh · [10]

Design a linear phase FIR filter using the Kaiser window for  $0.99 \leq |H(e^{j\omega})| \leq 1.01$  for  $0 \leq |\omega| \leq 0.3\pi$  and  $|H(e^{j\omega})| \leq 0.01$  for  $0.35\pi \leq |\omega| \leq \pi$ .

#### Step 1: read the ripples off the inequalities

- Passband  $1 \pm 0.01$ , so  $\delta_p = 0.01$ . Stopband  $\leq 0.01$ , so  $\delta_s = 0.01$ .

$$\delta = \min(\delta_p, \delta_s) = 0.01, \quad A = -20 \log_{10}(0.01) = \mathbf{40 \text{ dB}}$$

#### Step 2: the shape parameter

- $21 < A \leq 50$ , so the middle branch applies:

$$\beta = 0.5842(40 - 21)^{0.4} + 0.07886(40 - 21) = 0.5842(3.2474) + 1.4983 = \mathbf{3.3953}$$

#### Step 3: the length, and the window

- $\Delta\omega = 0.35\pi - 0.3\pi = 0.05\pi = 0.15708$ .

$$M = \frac{A - 8}{2.285 \Delta\omega} = \frac{32}{2.285(0.15708)} = 89.15 \Rightarrow M = 90, \quad \boxed{N = 91}$$

- $\omega_c = \frac{0.3\pi + 0.35\pi}{2} = 0.325\pi$ , and  $\alpha = 45$ .

$$w[n] = \frac{I_0\left(3.3953\sqrt{1 - \left(\frac{2n}{90} - 1\right)^2}\right)}{I_0(3.3953)}, \quad h[n] = h_d[n]w[n]$$

- **$\beta$  and  $N$  came from different specifications**, which is the whole advantage over a fixed window: the attenuation did not force a longer filter.

#### Every other Kaiser paper

**Eight of the ten share  $\delta = 0.01$** , so they share  $A = 40$  dB and  $\beta = 3.3953$  exactly. Only the transition width, and therefore  $N$ , changes. Learn one calculation.

| Papers        | Spec                                                     | $\delta$ | $A$   | $\beta$       | $N$ |
|---------------|----------------------------------------------------------|----------|-------|---------------|-----|
| 81 Ba, 81 Bh  | $0.3\pi / 0.35\pi, \delta = 0.01$                        | 0.01     | 40    | 3.3953        | 91  |
| 80 Bh         | $0.19\pi / 0.21\pi, \delta_p = 0.05$                     | 0.01     | 40    | 3.3953        | 225 |
| 80 Ba         | $0.016\pi / 0.08\pi$                                     | 0.01     | 40    | 3.3953        | 71  |
| 79 Bh         | $0.2\pi / 0.4\pi, \delta_p = 0.101$                      | 0.01     | 40    | 3.3953        | 25  |
| 78 Bh, 74 Ch  | band pass, $0.25\pi / 0.35\pi$<br>and $0.6\pi / 0.65\pi$ | 0.01     | 40    | 3.3953        | 47  |
| 75 Ash, 72 Ka | $0.19\pi / 0.21\pi$                                      | 0.01     | 40    | 3.3953        | 225 |
| 76 Ash        | $0.16\pi / 0.18\pi$                                      | 0.01     | 40    | 3.3953        | 225 |
| 73 Ch         | $0.09\pi / 0.14\pi, \delta_p = 0.02$                     | 0.01     | 40    | 3.3953        | 91  |
| 78 Ch         | $0.19\pi / 0.21\pi$ , both $\delta = 0.02$               | 0.02     | 33.98 | 2.6523        | 183 |
| 66 Ma         | $0.25\pi / 0.65\pi$ , both $\delta = 0.035$              | 0.035    | 29.12 | <b>1.9903</b> | 9   |

**66 Ma supplies a Bessel table, and it is mislabelled** The paper prints a table headed  $J_0(x)$  with values 1, 1.4826, 1.8926, 2.0508, 2.1675, 2.1718, 2.2642, 2.2796 at  $x = 0, 1.3165, 1.7237, 1.8455, 1.9271, 1.93, 1.9903, 2$ . **Every one of those eight is  $I_0(x)$** , the **modified** Bessel function, not  $J_0$ : the ordinary  $J_0$  oscillates and never exceeds 1, while these climb past 2. It is the right table with the wrong name. **And the table gives the answer away:**  $\delta = 0.035$  gives  $A = 29.12$  dB and  $\beta = \mathbf{1.9903}$ , which is one of its own  $x$  entries, so  $I_0(\beta) = 2.2642$  is supplied ready to use. The remaining entries are the  $x$  values needed for  $n = 0 \dots 8$  of a 9-tap window.

**73 Ch prints a passband that swallows its own stopband** It gives  $0.98 \leq |H| \leq 1.02$  for  $0 \leq \omega \leq 0.9\pi$ , then  $|H| \leq 0.01$  for  $0.14\pi \leq \omega \leq \pi$ . **The passband edge is larger than the stopband edge**, so the two overlap and no filter exists. Read  $0.9\pi$  as **0.09 $\pi$** : the transition  $0.09\pi$  to  $0.14\pi$  is then sensible and gives  $\Delta\omega = 0.05\pi$ ,  $\beta = 3.3953$  and  $N = 91$ . Note the misprint in a line and design the sensible filter.

**78 Bh and 74 Ch are band pass, not low pass** Their spec has three bands: stop up to  $0.25\pi$ , pass from  $0.35\pi$  to  $0.6\pi$ , stop from  $0.65\pi$ . **Use the narrower of the two transitions**, here both  $0.1\pi$ , so  $\Delta\omega = 0.1\pi$  and  $N = 47$ . The ideal response is the band-pass one,  $h_d[n] = \frac{\sin(\omega_2(n - \alpha)) - \sin(\omega_1(n - \alpha))}{\pi(n - \alpha)}$  with  $\omega_1 = 0.3\pi$  and  $\omega_2 = 0.625\pi$ , each edge again placed in the middle of its transition.

#### 4 The bookwork, which is a third of the chapter

**TOP 31/45** · Gibbs **TOP 8/45** and Remez **TOP 23/45** · **175 marks of Remez alone, and not one calculation**

- **Gibbs** is asked in eight papers for 3 to 6 marks. Three sentences: truncating with a rectangular window convolves the ideal response with a kernel whose sidelobes are only 13 dB down; the result overshoots the band edge by about 9 % with ringing either side; and **lengthening the filter does not reduce the overshoot**, only narrows the ripples, so the cure is a tapered window.
- **Remez** is asked in twenty-three, for 4 to 10 marks, and almost always wants **the flowchart**. Learn the seven boxes in §5.5 of the theory: input, initial guess of  $r + 2$  extremal frequencies, solve for  $\delta$ , interpolate and evaluate  $E(\omega)$  on a dense grid, exchange the extrema, test against the alternation theorem, output.
- The three "why is it better" marks: **it wastes no tolerance**, so it gives a **shorter filter** for the same spec; it allows **different passband and stopband ripple** through the weight  $W(\omega)$ ; and it ignores the transition band instead of constraining it.
- **Write  $\min_h \max_\omega |W(\omega)[H_d(\omega) - H(\omega)]|$  once** and the "mathematical expression" marks are collected.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full. → reads “leads to / therefore”, + reads “and”, = reads “is”. ✓ means the property holds, × that it does not.

| Short | Means                                           | Short | Means                                  |
|-------|-------------------------------------------------|-------|----------------------------------------|
| DSP   | digital signal processing                       | DSAP  | digital signal analysis and processing |
| DT    | discrete-time                                   | CT    | continuous-time                        |
| LTI   | linear time-invariant                           | LSI   | linear shift-invariant, the same thing |
| TI    | time-invariant                                  | BIBO  | bounded input, bounded output          |
| LCCDE | linear constant-coefficient difference equation | ROC   | region of convergence                  |
| DTFS  | discrete-time Fourier series                    | DTFT  | discrete-time Fourier transform        |
| CTFT  | continuous-time Fourier transform               | DFS   | discrete Fourier series                |
| DFT   | discrete Fourier transform                      | IDFT  | inverse DFT                            |
| FFT   | fast Fourier transform                          | DIT   | decimation in time                     |
| DIF   | decimation in frequency                         | MAC   | multiply and accumulate                |
| FIR   | finite impulse response                         | IIR   | infinite impulse response              |
| LPF   | low pass filter                                 | HPF   | high pass filter                       |
| BPF   | band pass filter                                | BSF   | band stop filter                       |
| BLT   | bilinear transformation                         | II    | impulse invariance                     |
| A/D   | analog to digital                               | D/A   | digital to analog                      |
| S/H   | sample and hold                                 | ZOH   | zero-order hold                        |
| SNR   | signal to noise ratio                           | PYQ   | past-year question                     |
| coeff | coefficient                                     | freq  | frequency                              |
| eqn   | equation                                        | no.   | number                                 |
| ops   | operations                                      | attr  | attribute                              |
| w/    | with                                            | w/o   | without                                |
| Eg    | for example                                     | Vs    | compare / differentiate                |
| +Fig  | a figure is wanted                              | (I)   | importance / need                      |
| (+)   | advantage                                       | (-)   | disadvantage                           |

## Exam paper codes

### Month codes

- **Ba** Baishakh **Jth** Jestha **Asa** Ashad
- **Shr** Shrawan **Bh** Bhadra **Ash** Ashwin
- **Ka** Kartik **Mng** Mangsir **Po** Poush
- **Ma** Magh **Ch** Chaitra
- **Ash and Asa are different months**: 2075 Ashwin and 2070 Ashad are separate papers.

### How a year code is written

- **Bold** = **Regular** exam. Plain = Back exam.
  - “New Back (2066 & Later Batch)” counts as Back.
  - A header printing “Regular / Back” in one box counts as Regular.
- Typewriter = the EX 753 paper (DSP, BEX, IV/II). Normal font = **CT 704** (DSAP, BEI/BCT, IV/I).
- The two axes are independent: **76 Ch** is CT 704 Regular, 76 Ash is CT 704 Back, 76 **Bh** is EX 753 Regular, 74 **Ma** is EX 753 Back.
- **69 Bh and 69 Bh** are two different papers, one in each course.

## Frequency chips

- **TOP n/45** asked in **10 or more** of the 45 papers.
- **HOT n/45** asked in **4 to 9** papers.
- **PIN n/45** asked in **1 to 3** papers.
- **verified/added** written or corrected against a reference because the source notes did not cover it, or covered it wrongly.
- **5** marks the question has actually carried. Several chips mean the same question has been set at different marks in different years.
- The count is over **45 papers**: 28 CT 704 (2066 Magh to 2082 Bhadra) and 17 EX 753 (2069 Bhadra to 2081 Chaitra).
- Counts come from the Detailed PYQ document, which reproduces every question in full.